

SMARTLIBRARY REQUIREMENTS SPECIFICATION

1. Overview

SmartLibrary is a desktop application for Limkokwing University's central library. It manages books, authors, members, loans, and book clubs. Backend uses Python OOP, PostgreSQL for storage, psycopg2 for connectivity, and PyQt5 for GUI.

2. Actors

- Librarian – full access to CRUD operations, loan workflows, book clubs.
- Member – limited access to view catalog and join book clubs.
- System – applies loan rules and constraints.

3. Use Cases

1. Login (role-based)
2. Manage Books (CRUD)
3. Manage Authors (CRUD)
4. Manage Members (CRUD)
5. Loan Workflow: borrow, return, overdue handling
6. Book Clubs: create, manage, add members
7. Search Catalog: search/filter books
8. Dashboard: statistics and most borrowed books

4. Functional Requirements

- Authentication with role-based access.
- CRUD operations for books, authors, members, clubs.
- Loan workflow enforcing rules: max 3 active loans, 7-day loan period.
- Search and filter operations on catalog.
- Audit log for operations.
- Dashboard KPIs.

5. Non-Functional Requirements

- Performance: <2 seconds response.
- Usability: clean UI with PyQt5.
- Security: password hashing, RBAC.
- Availability: reliable PostgreSQL backend.
- Maintainability: modular Python OOP code and DAOs.

6. Wireframes (Textual)

- Login Screen: email, password, role selector.
- Dashboard: KPIs, navigation buttons.
- Book Catalog: search bar, filters, list view.
- Loan Management: issue/return books.
- Book Club Module: list clubs, members, add/remove.